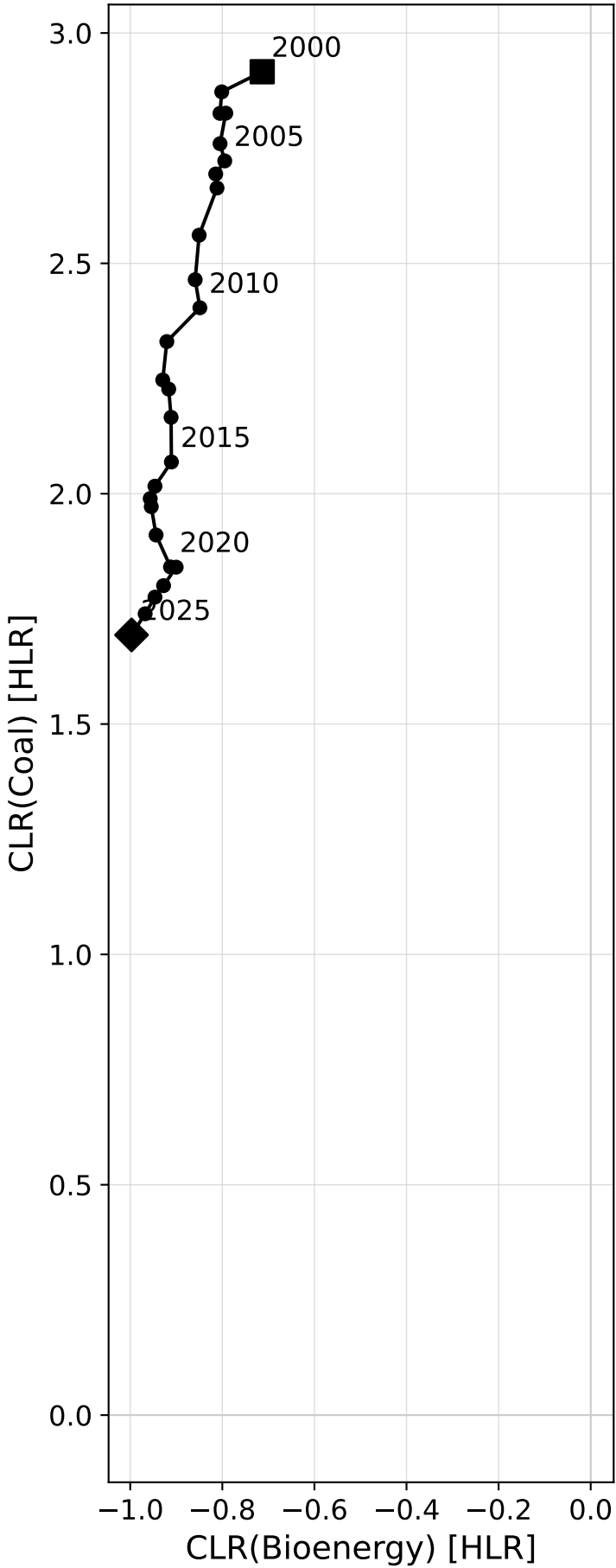
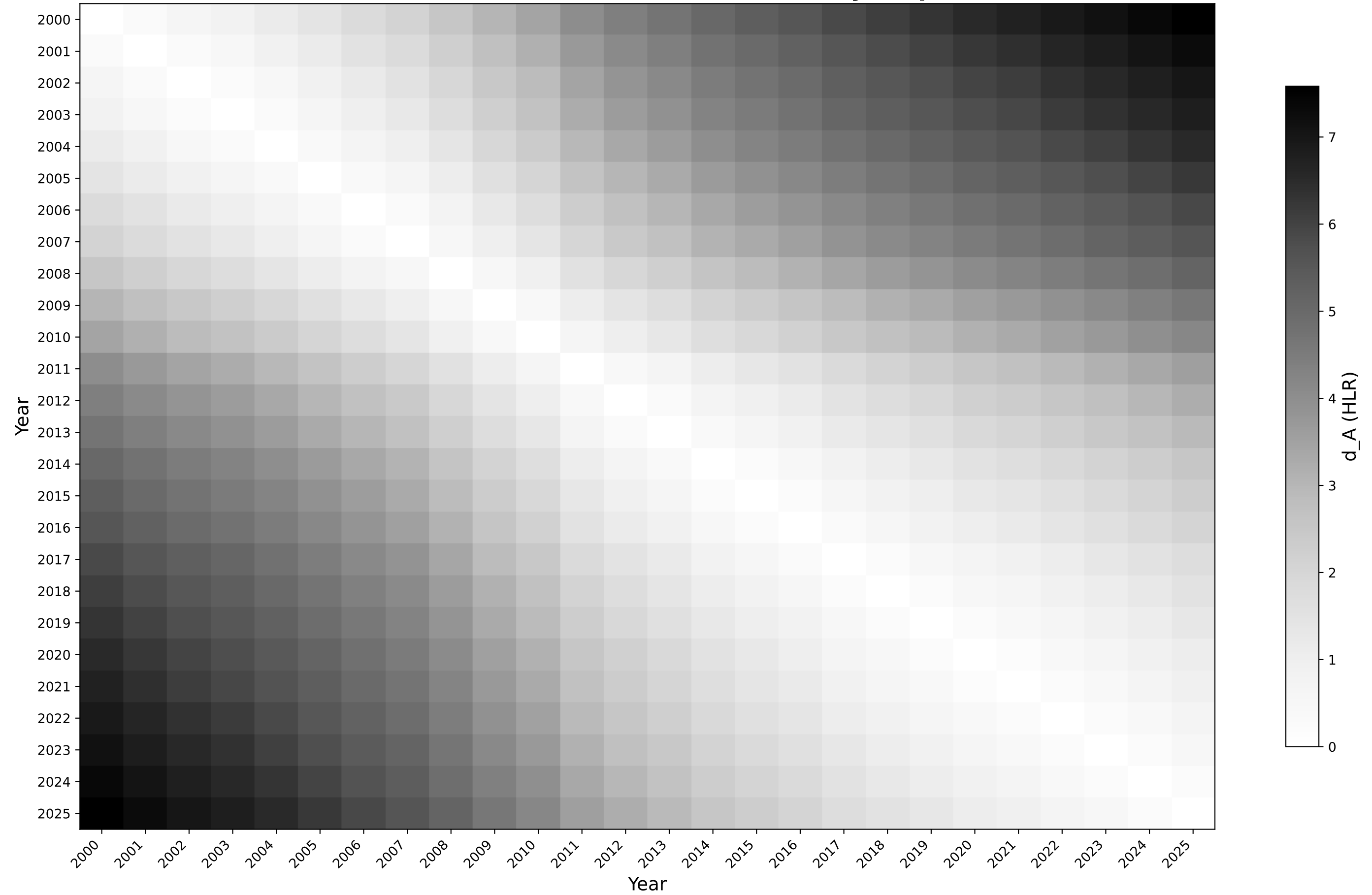


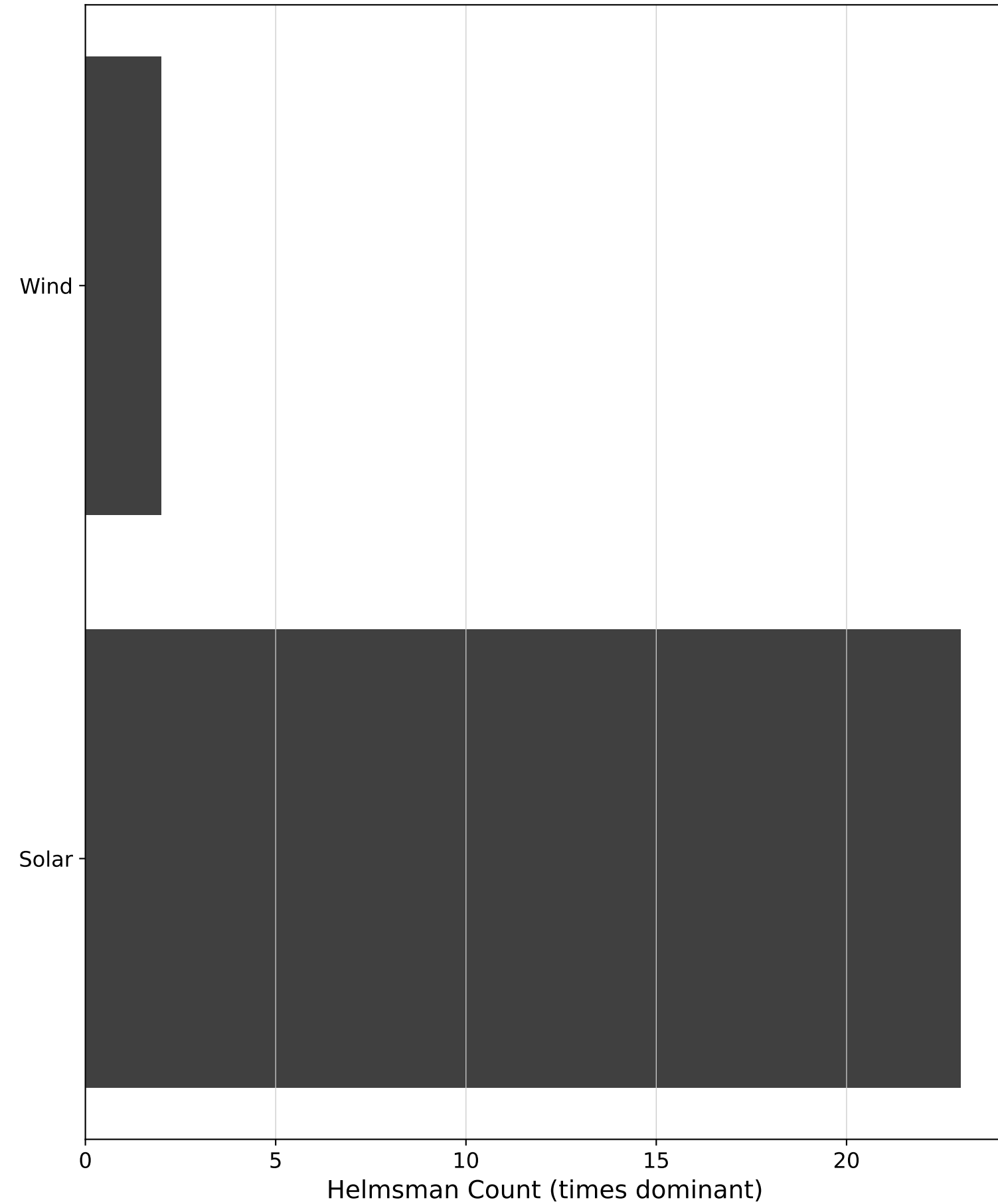
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



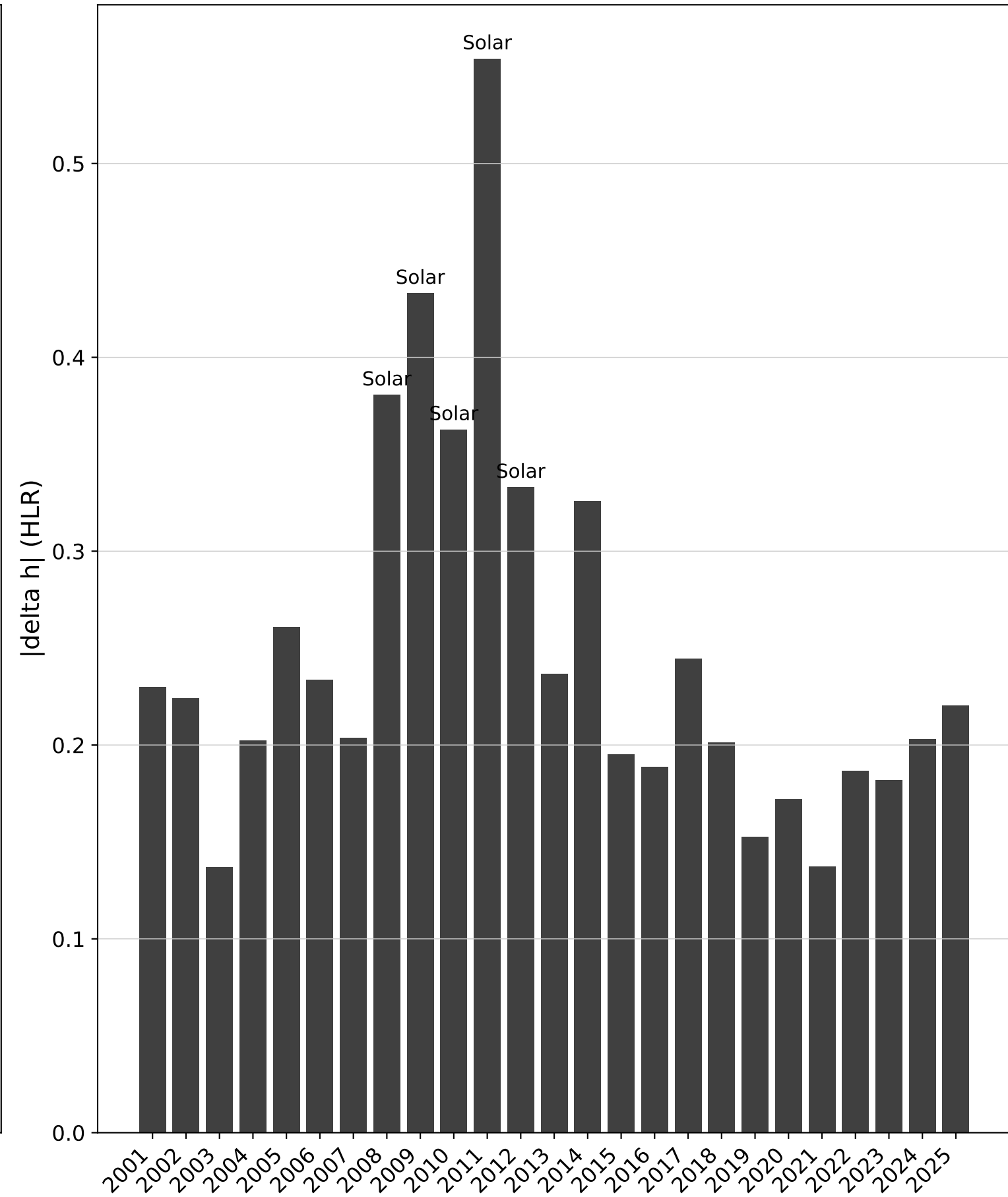
STAGE 2 — Aitchison Distance Matrix (all year pairs)



HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



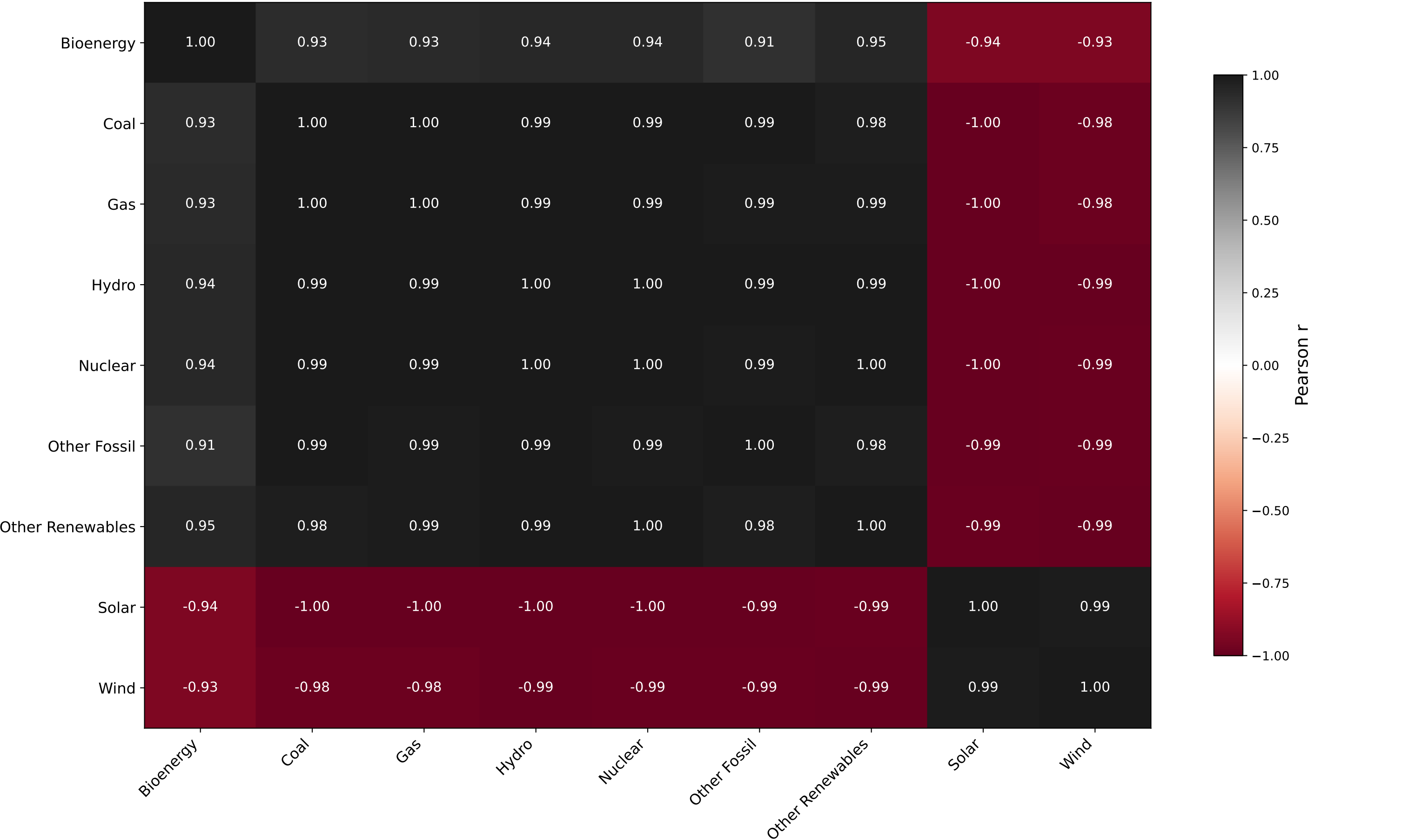
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2000	2025	7.5841	67.2	Solar
2	2000	2024	7.3480	63.8	Solar
3	2001	2025	7.2851	65.9	Solar
4	2000	2023	7.1308	60.8	Solar
5	2001	2024	7.0491	62.6	Solar
6	2002	2025	7.0156	64.8	Solar
7	2000	2022	6.9229	57.8	Solar
8	2001	2023	6.8318	59.5	Solar
9	2003	2025	6.8097	63.8	Solar
10	2002	2024	6.7784	61.4	Solar
11	2000	2021	6.7029	54.8	Solar
12	2001	2022	6.6242	56.6	Solar
13	2003	2024	6.5719	60.5	Solar
14	2002	2023	6.5598	58.4	Solar
15	2000	2020	6.5308	52.3	Solar

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2000	2007	2025	1.9693	3.64
2	2000	2008	2025	1.8595	3.00
3	2000	2007	2024	1.8505	3.52
4	2001	2007	2025	1.7748	4.08
5	2000	2007	2023	1.7414	3.42
6	2000	2008	2024	1.7317	2.91
7	2000	2006	2025	1.6982	4.26
8	2000	2009	2025	1.6787	2.50
9	2001	2008	2025	1.6760	3.27
10	2000	2007	2022	1.6689	3.32

STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2000	0.2874	Hs-2	66.7	0.00	0.0000	---	14979
2001	0.2886	Hs-2	62.9	1.48	0.3185	Solar	11398
2002	0.2890	Hs-2	59.4	1.59	0.3117	Wind	9331
2003	0.2932	Hs-2	56.8	1.17	0.2274	Solar	7881
2004	0.2911	Hs-2	53.6	1.76	0.3178	Wind	6264
2005	0.2920	Hs-2	50.0	1.68	0.3269	Solar	4591
2006	0.2960	Hs-2	46.6	2.00	0.3442	Solar	3476
2007	0.2979	Hs-2	43.6	1.85	0.3086	Solar	2704
2008	0.2922	Hs-2	38.9	2.51	0.4644	Solar	1696
2009	0.2879	Hs-2	34.1	3.20	0.5200	Solar	1006
2010	0.2858	Hs-2	31.0	3.38	0.4302	Solar	660
2011	0.2811	Hs-2	26.8	5.26	0.6288	Solar	360
2012	0.2738	Hs-2	24.5	3.70	0.4012	Solar	358
2013	0.2716	Hs-2	22.9	3.12	0.3089	Solar	360
2014	0.2657	Hs-2	21.0	3.72	0.3689	Solar	342
2015	0.2529	Hs-2	19.7	2.80	0.2619	Solar	328
2016	0.2466	Hs-2	18.9	3.01	0.2460	Solar	325
2017	0.2418	Hs-2	18.1	4.06	0.3202	Solar	323
2018	0.2391	Hs-2	17.7	3.44	0.2588	Solar	324
2019	0.2289	Hs-2	17.3	2.82	0.2111	Solar	323
2020	0.2178	Hs-2	16.7	3.36	0.2533	Solar	310
2021	0.2120	Hs-2	16.6	2.86	0.2037	Solar	327
2022	0.2025	Hs-2	16.3	3.30	0.2367	Solar	329
2023	0.1950	Hs-2	16.3	3.04	0.2140	Solar	336
2024	0.1868	Hs-2	16.6	3.22	0.2303	Solar	350
2025	0.1771	Hs-2	16.7	3.44	0.2455	Solar	356